Personal Appraisal

Module: Group Project
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1. Introduction

This is my personal assessment of the group project, Deep Learning for Speech Recognition: An ASR Challenge with Modern Models. The particular project tried to analyse the different performing systems during this type of task at different architectures and datasets, with cross-domain experimentation, in order to have an understanding of what challenges the systems will encounter at this stage.

It involved experimentation on several models (listed below), including Whisper, Conformer, XLSR and Quartznet-based architecture using datasets like Mozilla Common Voice and LibriSpeech. It was challenged to analyse the performance of models on various metrics (Word Error Rate(WER), Character Error Rate(CER) and so on), design experiments for different methods and validate results from the experiments.

In this evaluation, I will review the overall management and output of the project, also what came to be from working in a team environment, what was done well and what can be improved with regards to my personal role regarding design implementation, experimentation and documentation.

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2. Reflection on the Overall Project

2.1 Project Organisation and Management

One thing that worked well was weekly meetings of check-ins on how progress was going. During the early stages of the project, several research paths and datasets were investigated before finally deciding to focus primarily on speech-to-text recognition. The group adopted a more collaborative approach — members would each attempt their hand at different architectures, which were then compared later.

Work has been split into micro-tasks such as:

- dataset exploration
- literature review
- model experimentation

- evaluation and result analysis
- report writing

The models themself -- Conformer, XLSR, etc. were divided into sub-groups for working on them in parallel. This proved advantageous to help divide the work and still let members concentrate on smaller pieces of a bigger project.

There were regular meetings to track progress, identify what was (and wasn't) working and change the trajectory of the work. The minutes provided transparency and a precise record of who shouldered responsibility for any particular action.

2.2 Successes of the Project

One of the major successes of this project was to participate in applying and evaluating many state-of-the-art ASR architectures on several datasets. This enabled us to build end-to-end pipelines covering data preparation, training, evaluation and interpretation of the results.

Another key success was model comparability, allowing us to see how distinct architectures perform against similar data. Some worked well (the Whisper-based model → compared relatively low WER and CER values.

This included testing the model trained on one dataset with data from a different dataset (cross-domain). It helped raise awareness of the challenges of generalisation for speech recognition systems.

Also, this paper explores several multilingual speech recognition systems using different datasets, like the Omnilingual dataset, which revealed an enormous scope at a large scale for Multi-Linguistic ASR models.

A further key to their success was the collaborative environment of the group. During the process, team members validated findings with each other regularly and partnered to help troubleshoot technical issues that arose in implementing models.

2.3 Difficulties and Challenges

Despite being successful, the project suffered from some setbacks.

The largest problem was not being able to test it on a clustered environment. Since training large deep learning models is resource-intensive, the amount of hardware needed limited how far experimentation could be tried. Some models needed to be trained on fewer epochs or reduced datasets.

The last, but not least, challenge has been the complexities of ASR pipelines. Building and debugging speech recognition models ain't simple work; several steps are involved in this: data preprocessing,

feature extraction, model training and evaluation. Errors at any stage could widely affect subsequent output.

The cross-domain experiments also demonstrated that when a model performed well on one dataset, there was often little positive performance on another dataset during training. This underlined the difficulty in developing ASR systems that generalise consistently across different recording conditions.

2.4 Tools, Methods and Approaches

Over the entire process, several modern tools were used, including deep learn189ing libraries and speech processing toolkits. We had tried using Whisper and Conformer-based architectures, as well XLSR based architecture on publicly available speech datasets.

Word Error Rate (WER) and Character Error Rate (CER) were the most common metrics that reported results for evaluating the performance of speech recogniser research.

The team went on to experiment with different architectures and datasets, which helped them understand some of the pros and cons of various ASR approaches in practice.

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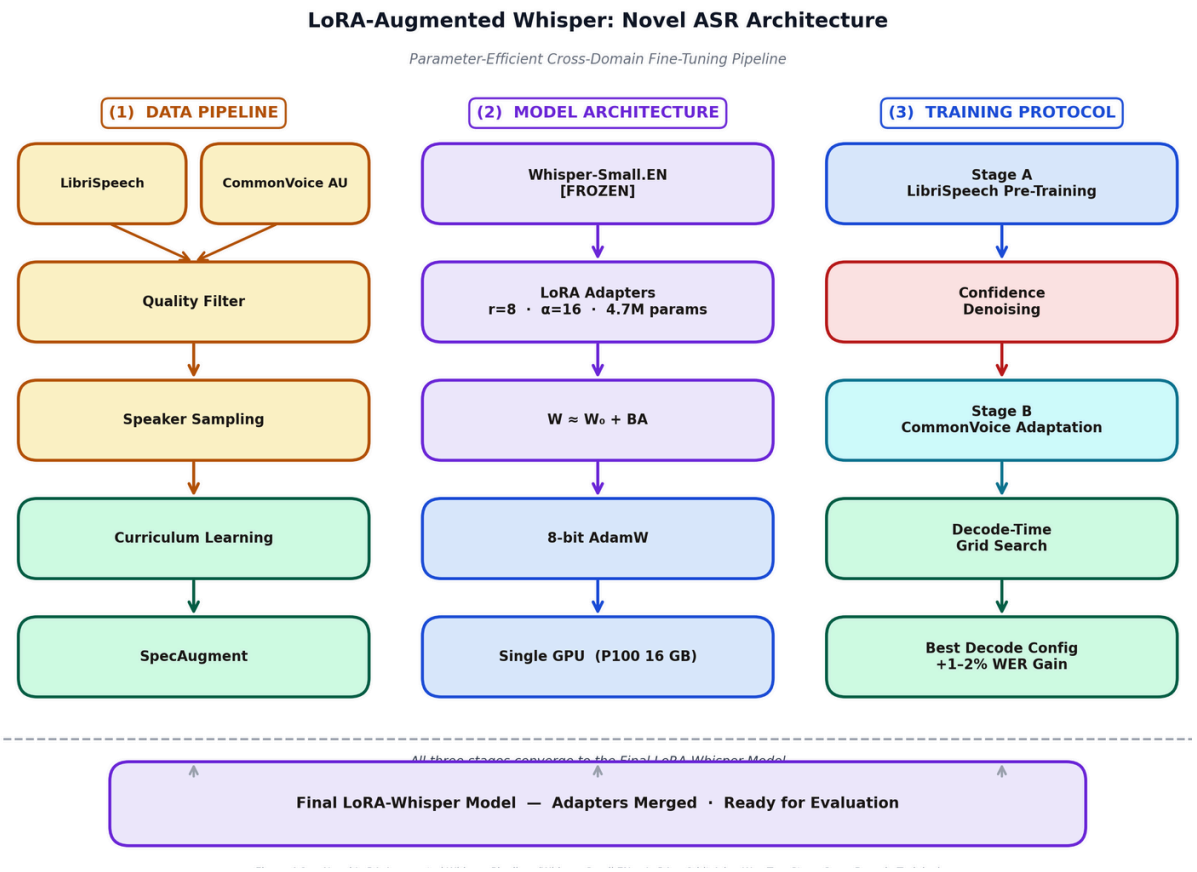
3. Reflection on My Individual Contribution

3.1 Design

Furthermore, my focus as a designer was to maintain the end-to-end nature of cross-domain ASR while remaining research-grade and utilising hardware capabilities to their fullest. I created baseline models for the Whisper small model on Mozilla Common Voice. I have also used a two-stage sequential pipeline: Stage A fine-tuning on LibriSpeech for broad acoustic representations and then Stage B domain adaptation using Mozilla Common Voice Australian English (en-AU).

Finally, I proposed a new ASR Architecture: LoRA Augmented Whisper Model. Below are the proposed and implemented architectures. This pipeline was made unique compared to a plain fine-tune baseline with the following four novel design innovations: (quality-aware curriculum, speaker-balanced sampling, confidence-based denoising, on-the-fly collation)



3.2 Implementation

I separated my implementation into four standalone notebooks, one for each ASR pipeline component:

Notebook A — Unified EDA + Preprocessing (Common Voice+ LibriSpeech)In this notebook, we merged Common Voice metadata and created a common dataframe with absolute audio path visible, walking through an overview of the complete dataset.

Notebook B — whisper-small-mcv. ipynb (Common Voice Full Fine-Tuning). Now the notebook performed a complete fine-tune pipeline for openai/whisper-small on Mozilla Common Voice en-AU that is GPU efficient.

Notebook C — cross-domain-asr-pipeline. ipynb (Cross-Domain Evaluation + HPO-Lite). This notebook built an organised cross-domain experiment where Whisper-Small was trained on LibriSpeech (Stage A), then zero-shot evaluated on Common Voice en-AU to quantify domain shift and finally adapted with Stage B conditioning.

Notebook D — Whisper_LORA_Cross_Domain. ipynb (LoRA). This notebook ran the complete end-to-end Novel LoRA-augmented cross-domain pipeline: Whisper-Small with LoRA adapters (rank 8, both q_proj/v_proj) trained on LibriSpeech and fine-tuned to Common Voice en-AU, all four design innovations active simultaneously.

3.3 Integration and Collaboration

The individual EDA and preprocessing framework I had set up — the unified Common Voice + LibriSpeech pipeline, including integrity checks, speaker-imbalance Gini/Lorenz analysis and audio quality diagnostics — served a common analytical foundation that multiple other group members built upon in their own implementations. I used outputs from EDA in determining my own preprocessing strategy, and many other teammates did the same, so while I wrote Notebook A, it contributed to a multiplicative effect beyond just my single pipeline!

Finally, I did get to interact with my teammates from across the sub-hub. I brought several tips about cross-domain evaluation paradigm setup strategies (e.g. config), how to load datasets in Kaggle environments & avoid the pitfalls + LoRA config best practices for related architectures. All four of my notebooks feature a modular structure for reuse in the group and publication.

3.4 Documentation and Administration

All four notebooks were created with reproducibility and publication-ready reporting in mind. All the notebooks were well structured, compartmentalising tasks into named code sections, interspersed with inline comments to explain non-obvious implementation choices and export routines to give clean artefacts. These outputs were in a format that could be plugged and played directly into the group report with no post-processing required. I also wrote the meat of the report, that is what I mentioned above: Related Works, Dataset Collection and Methodology.

Side note, for the contribution aspects on group project at administrative level, I maintained stable traction as a leader contributor, directly engaging with EDA, full fine-tuning and LoRA tuning in addition to crossing domains with our pipeline. I ensured that my deliverables were sufficiently complete to be good reference implementations:— they generally included a cross-domain pipeline covering everything from raw data loading all the way through to final error analysis export.

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4. Personal Learning and Reflection

This project drew upon the widest range of applied machine learning experience I had gained to that point, requiring me to simultaneously coordinate skills in data engineering, model training, evaluation methodology and research communication. The biggest technical takeaway was working with real hardware constraints.

Cross-domain evaluation results gave valuable insights into how performance on the training set differs from real-world generalisation. The zero-shot WER on Common Voice en-AU — an order of magnitude higher than performance on the Stage A LibriSpeech dev set — crystallised the domain shift problem in a way writing about it did not. The challenge of cross-domain generalisation in ASR

— how to build systems that transfer reliably across acoustic regimes, accents and recording conditions without requiring large amounts of labelled data from the target domain in which to train them — remains an important unsolved scientific problem with immediate practical consequences.

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5. Contribution Ranking

Team Member	Contribution Score (1-10)
Soham Dharne (2500908)	8
Tarak Hossain (2500810)	6
Uswa Mahmood (2501194)	6
Sayed Hossain Bhuiyan (2500958)	5
Edward Hoogewerf (2501806)	6
Aaditya Nair (2500803)	7
Latha Mahendra (2508157)	6
Aniket Domadiya (2501463)	8